

Puerto Rico Reasonable Potential (RP) Calculator

User Guide

Contents

1	Overview	3
1.1	What the App Does	3
1.2	What the App Does Not Replace	3
2	Regulatory Background	4
2.1	Reasonable Potential Under the Clean Water Act	4
2.2	The Puerto Rico Water Quality Standards Regulation	4
2.3	Water Classifications Relevant to NPDES Permitting	5
3	Getting Started	6
3.1	Two Workflows at a Glance	6
3.2	Navigating the App	6

NPDES Reasonable Potential (RP) Calculator

[User Guide](#)

Select Analysis Workflow



Calculate RWC Using NPDES Data

[Connect to ICIS-NPDES](#)

Choose this if you have the NPDES ID. You will have the option to add local data if needed.



Calculate RWC Using Local Data

[Standalone Calculator](#)

Choose this if no NPDES ID exists but you have data to use.

3.3

7

4	NPDES Workflow	7
4.1	Step 1: Search and Select a Facility	7
4.2	Step 2: Configure Filters	8
4.2.1	NPDES Forms	9
4.2.2	Water Classifications	9
4.3	Step 3: Fetch DMR Data	10
4.4	Step 4: Review the Data Overview	11

4.5	Step 4b: Resolve Parameters Needing Attention	13
4.6	Step 5: Configure Analysis Settings	13
4.6.1	Dilution Ratio	14
4.6.2	Confidence Level and Target Percentile	14
4.6.3	Hardness Settings (Class SD only)	15
4.6.4	pH and Temperature (for TAN, Class SD only)	15
4.7	Step 6: Append Local DMR Data (Optional)	16
4.8	Step 7: Run the RP Analysis	16
5	Standalone Workflow	17
5.1	Step 1: Open the Standalone Calculator	17
5.2	Step 2: Download and Complete the DMR Template	17
5.3	Step 3: Upload and Configure	17
5.4	Step 4: Run the RP Analysis	17
6	Findings Page	18
6.1	Outfall and Pollutant Selection	18
6.2	Quick Stats Panel	18
6.3	Interactive Inspector	20
6.3.1	Observed vs. WQS / RWC Plot	20
6.4	RP Summary Table	21
7	Report Generation	22
7.1	Download Options	22
7.2	Report Contents	23
8	Methodology	24
8.1	The Statistical Projection Approach (TSD Section 3.3.2)	24
8.1.1	The Multiplying Factor	24
8.1.2	The Receiving Water Concentration	25
8.1.3	Worked Example	26
8.2	Hardness-Dependent Metal Criteria	26
8.3	Total Ammonia Nitrogen (TAN)	27
8.4	pH	28
8.5	Temperature	28
9	Interpreting Results and Common Cases	29
9.1	Multiple Water Classes Selected	29
9.2	Metals with Hardness = Range (Class SD)	29
9.3	Unit Conversion Issues	29
9.4	No Data for a Parameter	30
9.5	Prior Limits in the Quick Stats Panel	30
10	Appendix A: NPDES Application Forms and Pollutants	30
11	Appendix B: Technical References	31
12	Appendix C: Glossary	31

1 Overview

The **Puerto Rico Reasonable Potential (RP) Calculator** is a web-based application designed to support EPA Region 2 permit writers in determining whether reasonable potential (RP) exists for an NPDES-permitted discharge to cause or contribute to an exceedance of Puerto Rico's water quality standards (WQS). The application automates the most data-intensive steps of the RP analysis: retrieving effluent monitoring data from a local copy of EPA's ICIS-NPDES database (refreshed on a weekly schedule), aligning those data with the applicable Puerto Rico WQS, performing the statistical projection described in the EPA Technical Support Document for Water Quality-Based Toxics Control (TSD, 1991), and generating a documented, reproducible PDF report.

The application is scoped to **Puerto Rico NPDES permits** and references the **Puerto Rico Water Quality Standards Regulation (PRWQSR), as amended April 2025** (DNER Resolution R-25-006). All numeric criteria, water classifications, and hardness-dependent metal limits applied by the app are drawn directly from that regulation.

1.1 What the App Does

For each pollutant and outfall combination at a selected facility, the app:

1. Retrieves maximum-statistic DMR values from the local ICIS-NPDES database (and, for pH, the minimum statistic as well), or accepts locally supplied data
2. Converts reported units to WQS units where necessary
3. Calculates a **Receiving Water Concentration (RWC)** using the statistical multiplying factor (MF) method from TSD Section 3.3.2
4. Compares the RWC to the applicable criterion from the PRWQSR for each selected water classification
5. Returns a YES / NO / DEPENDS RP determination for each pollutant $\times$ outfall $\times$ water class combination
6. Generates a PDF report — and optionally a ZIP package with supporting data — that documents all inputs, settings, and findings

1.2 What the App Does Not Replace

The app is an analytical aid, not a substitute for permit writer judgment. It does not:

- Perform whole effluent toxicity (WET) analysis
- Calculate waste load allocations (WLAs) or mixing zone analyses
- Account for background receiving water concentrations
- Evaluate narrative water quality criteria
- Replace the fact sheet or other permit development documentation requirements

When the app returns a YES RP finding, the permit writer must still determine the appropriate effluent limitation based on the applicable water quality standard, the dilution ratio, and any mixing zone authorization. When it returns NO, that finding is based on the statistical projection alone and should be considered alongside other factors described in 40 CFR 122.44(d)(1).

2 Regulatory Background

2.1 Reasonable Potential Under the Clean Water Act

Under 40 CFR 122.44(d)(1)(i), NPDES permits must include effluent limits for pollutants that are or may be discharged at levels that will cause, have the reasonable potential to cause, or contribute to an excursion above any applicable water quality standard. Determining whether a discharge has “reasonable potential” is therefore a required step in permit development for any parameter that is monitored at an outfall.

EPA’s 1991 Technical Support Document for Water Quality-Based Toxics Control (TSD) provides the recommended methodology for making this determination using effluent monitoring data. The core question is whether the upper bound of the expected effluent concentration distribution, projected into the receiving water accounting for dilution, could exceed the applicable criterion. The statistical approach in the TSD uses the coefficient of variation (CV) of the effluent data and the number of samples to construct this projection without assuming a specific distributional form for the effluent — though the lognormal distribution is used as the basis for the multiplying factors.

2.2 The Puerto Rico Water Quality Standards Regulation

The PRWQSR, administered by Puerto Rico’s Department of Natural and Environmental Resources (DNER), establishes the water quality standards that NPDES permits in Puerto Rico must protect. The regulation classifies Puerto Rico’s waters by designated use and sets numeric criteria for a wide range of pollutants. The 2025 amendment (Resolution R-25-006, effective June 27, 2025) is the currently applicable version.

The standards that drive RP determinations fall into three main categories as implemented in the app:

- **Fixed numeric criteria** — a single concentration limit for a given pollutant and water classification (e.g., arsenic in Class SB at 36 µg/L)
- **Hardness-dependent criteria** — applicable to seven metals in Class SD waters; the criterion is computed from the receiving water hardness using regression equations specified in Rule 1303.1(J)(1) (see Section 8.2)

- **Condition-dependent criteria** — applicable to Total Ammonia Nitrogen (TAN) in Class SD waters; the criterion varies with receiving water pH and temperature as specified in Rule 1303.2(C)(2)(1) (see Section 8.3)

2.3 Water Classifications Relevant to NPDES Permitting

The PRWQSR classifies Puerto Rico’s waters into several categories. The app includes the four classifications most relevant to NPDES point source discharges. Understanding these classifications is important because the applicable criterion — and therefore the RP determination — differs by classification.

Class SB — Coastal and Estuarine Waters Class SB encompasses all coastal and estuarine waters of Puerto Rico not designated as the more protective Class SA (which includes the bioluminescent lagoons and other waters of exceptional ecological value where point source discharges are not permitted). Class SB extends from mean sea level to a maximum of approximately 10.35 miles offshore and includes associated wetlands. Designated uses include primary and secondary contact recreation, and propagation and maintenance of desirable species including threatened and endangered species. Most NPDES-permitted facilities discharging to coastal waters in Puerto Rico discharge to Class SB receiving waters.

Class SD — Surface Waters Class SD covers all surface waters of Puerto Rico — streams, rivers, lakes, reservoirs, and inland water bodies — except those designated as the more protective Class SE (Laguna Tortuguero and Laguna Cartagena). Designated uses include use as a raw source of public water supply, propagation and maintenance of desirable species, and primary and secondary contact recreation. Class SD criteria are generally more stringent than Class SB for most parameters because of the drinking water supply designation. Several criteria under Class SD depend on receiving water hardness (for metals) or on pH and temperature (for TAN). In addition, three parameters — total nitrogen, total phosphorus, and selenium — carry **separate criteria depending on whether the receiving water is a stream or a reservoir/lake**. The app reports these as distinct water-class lines (e.g., *class SD waters — stream* vs *class SD waters — reservoir/lake*) so the permit writer can see which receiving-water type drives a finding.

Class SG — Ground Waters Class SG includes all ground waters of Puerto Rico, including those that discharge into coastal, surface, and estuarine waters. Designated uses are drinking water supply and agricultural use including irrigation. Ground water criteria are typically the drinking water (DW) or human health (HH) values, whichever is more stringent given the receiving water body into which the ground water flows.

Surface Waters (general) In the app’s water type filter, “Surface Waters” maps to the PRWQSR’s general surface water standards (Rule 1303.1) that apply across all surface water classifications, as distinct from the class-specific criteria under Rule 1303.2.

i Note

Selecting water classifications in the app: The water classification selection in the app determines which WQS rows from the crosswalk are active for both data filtering and RP comparison. Selecting multiple classifications will produce RP determinations for each class independently — the app reports a “worst-case” RP across all selected classes, as well as the per-class breakdown. When in doubt about which classification applies to a receiving water body, consult the PRWQSR classification maps (Rule 1302) or contact the DNER Water Quality Program.

3 Getting Started

3.1 Two Workflows at a Glance

The app offers two entry points depending on whether the facility appears in the local ICIS-NPDES database.

	NPDES Workflow	Standalone Workflow
Best for	Existing permitted facilities	New permits, reissuances with no prior ID, or supplementing the database with local measurements
Data source	Local ICIS-NPDES database (weekly refresh)	User-supplied CSV using the app’s template
Requires	NPDES permit ID	Completed DMR template
Can supplement with local data	Yes — via the Append DMR function	N/A — local data is the primary source

Both workflows share the same Analysis Settings, Findings, and Report pages. The only difference is how effluent data are loaded.

3.2 Navigating the App

The app uses a single-page layout with page routing. A persistent top bar appears on every page with a ← **Back to Start** button (returns to the landing page) and a **User Guide** button (opens this document in a modal window). Within each page you also move using contextual buttons (e.g., “← Back to Search,” “← Back to Settings,” “Return Home”).

Select Analysis Workflow


Calculate RWC Using NPDES Data
[Connect to ICIS-NPDES](#)
Choose this if you have the NPDES ID. You will have the option to add local data if needed.


Calculate RWC Using Local Data
[Standalone Calculator](#)
Choose this if no NPDES ID exists but you have data to use.

3.3

4 NPDES Workflow

Use this workflow when the facility has an active NPDES permit and effluent monitoring data available in the local ICIS-NPDES database.

4.1 Step 1: Search and Select a Facility

Click **Connect to ICIS-NPDES** on the landing page. This opens the facility selection page, which loads the list of NPDES permits for Puerto Rico from the local ICIS-NPDES database at application startup.

Use the **Search Permit ID or Facility Name** box to find the facility of interest. You can type either the NPDES ID (e.g., PR0001031) or any part of the facility name — the search is case-insensitive and matches partial strings. Select the facility from the dropdown.

Once a facility is selected:

- The **map** displays a marker at the facility's reported latitude/longitude from the facility record in the local database
- The **Facility Details** table displays key permit attributes from the facility record in the local database
- The **Settings** panel and **Fetch DMR Data** button become visible

NPDES Reasonable Potential (RP) Calculator

[User Guide](#)

Search Permit ID or Facility Name

PR0023451 - VIATRIS PHARMACEUTICALS LLC

Start Date

2021-04-15

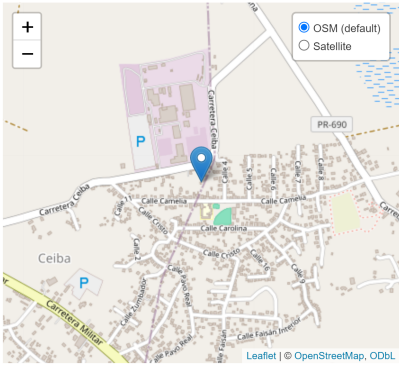
End Date

2026-04-15

Return Home

OSM (default)

Satellite



Facility Details

	SourceID	CWPName	CWPState
1	PR0023451	VIATRIS PHARMACEUTICALS LLC	PR

Settings

NPDES Forms

Water Classifications

Water Type

Fetch DMR Data

Figure 1: Facility selection page showing the search box, map with marker, and Facility Details table

Set the **date range** for DMR retrieval using the Start Date and End Date inputs. The default range is the five years prior to today's date, which is appropriate for most permit reissuances. The date range controls which DMR records are pulled from the database — it does not filter the facility list.

Note

Selections persist within your session: If you navigate away from the search page and return — for example, to revise a search — your permit ID, date range, NPDES forms, and water classification are retained. These reset only when the app is restarted. Analysis Settings on the Summary page (hardness, pH, temperature, dilution ratio, confidence level) are not persisted, because they re-select naturally when a different permit is fetched.

4.2 Step 2: Configure Filters

Before fetching data, configure the two filters that control which parameters and WQS criteria are included in the analysis.

4.2.1 NPDES Forms

Select one or more **NPDES application forms** from the multi-select dropdown. NPDES applications require permittees to report specific pollutants depending on the type of facility and the nature of the discharge. The forms available in the app correspond to the standard EPA application forms:

- **Form 2A** — New and existing publicly owned treatment works (POTWs)
- **Form 2B** — Concentrated Animal Feeding Operations (CAFOs)
- **Form 2C** — Existing manufacturing, commercial, mining, and silviculture facilities
- **Form 2D** — New manufacturing, commercial, mining, and silviculture facilities
- **Form 2E** — Facilities that do not discharge process wastewater

The form selection serves two purposes. First, it defines the **universe of pollutants** that are expected to be reported for the facility — this determines which pollutants are highlighted in the Data Overview (Coverage Summary) table. Second, it controls **display**: the form association is shown in the Coverage Summary’s **Form** column and in the report’s *Reason Included* line. Form selection no longer removes parameters from the analysis — any parameter with DMR data and an applicable WQS criterion is evaluated for RP regardless of form association. Parameters not associated with a selected form are labeled *Not in selected forms* in the Form column but are still included in the calculation.

Tip

Selecting the right form: If you are unsure which form applies, refer to the facility’s permit application or its permit record. Most municipal wastewater treatment plants use Form 2A. Most industrial and manufacturing dischargers use Form 2C or 2D. When multiple forms apply (e.g., a POTW receiving industrial pretreatment discharges), select all applicable forms.

4.2.2 Water Classifications

Select one or more **water classifications** from the multi-select dropdown. The choices correspond to Puerto Rico’s PRWQSR classifications described in Section 2.3:

Selection	Maps to PRWQSR Classification
All	All classes in the crosswalk
Class SB	Class SB coastal and estuarine waters
Class SD	Class SD surface waters (including drinking water subclass)
Class SG	Class SG ground waters
Surface Waters	General surface water standards

These selections determine which WQS criteria rows are active for comparison. When multiple classifications are selected, the app performs the RP comparison independently for each class and reports the worst-case result. Select **All** to evaluate against every applicable standard in the crosswalk — this is appropriate when the receiving water classification is uncertain or when you want to be comprehensive.

! Important

Class SD and hardness-dependent criteria: If **Class SD** is selected, the conditional receiving water inputs panel will appear (see Section 4.6.3). You will need to provide a hardness value or choose the range analysis option before running RP. This panel is only shown for Class SD because the hardness-dependent metal criteria and the TAN criterion are specific to Class SD surface waters under the PRWQSR.

4.3 Step 3: Fetch DMR Data

Click **Fetch DMR Data**. The app queries the local ICIS-NPDES database for effluent monitoring records matching the selected permit ID and date range, then applies the following processing steps automatically:

NODI code handling: Records with NODI codes B (“no discharge”) or Q (“no discharge this period”) are set to a value of zero. This is consistent with the approach in the TSD — a period of no discharge represents a zero concentration and should be included in the sample set. Records with other NODI codes that prevent a numeric value from being reported are excluded.

Unit detection and filling: DMR reports occasionally omit unit labels for individual records, particularly for NODI-coded rows. The app attempts to fill missing units by: (1) using the most frequently reported unit for that parameter across all records from the permit, and (2) falling back to the WQS unit hint from the crosswalk if all records for the parameter have missing units.

Unit conversion: Each reported value is checked against the WQS unit for its parameter. The converter handles concentration conversions ($\mu\text{g/L} \leftrightarrow \text{mg/L}$, $\text{ng/L} \rightarrow \mu\text{g/L/mg/L}$, $\text{ppm} \rightarrow \text{mg/L}/\mu\text{g/L}$), recognizes several units that are written differently but mean the same thing (e.g., ICIS long-form bacteria and color units), and converts Fahrenheit to Celsius for temperature. Mass-load and flow units (e.g., kg/d , MGD, Cubic Meters per Day) are **excluded** from the WQS comparison by design rather than treated as errors. When a parameter has no WQS criterion for the selected water class, its records are flagged **NO WQS**. A genuinely unrecognized unit is flagged **FAIL** and the value is retained at face value. Every status is shown in the Coverage Summary and carried into the report. See the Unit Status table in Step 4.

After a successful fetch, the app navigates to the **Summary (Data Overview)** page.

4.4 Step 4: Review the Data Overview

The Data Overview page displays the **Coverage Summary** table, which lists the pollutants relevant to this analysis — those associated with the selected NPDES forms plus any additional parameters that have DMR data — alongside a count of available DMR samples and a unit conversion status indicator. Parameters not associated with a selected form are marked *Not in selected forms* in the Form column.

NPDES Reasonable Potential (RP) Calculator

[User Guide](#)

Analysis Settings
NPDES Forms: 2C, 2F

Dilution Ratio

Confidence Level

Target Percentile (upper bound)

Hardness Setting
☒ Use range
☐ Set Hardness

Metal Limit Settings
pH
☒ Use DMR
☐ Set pH
Temperature (C)
☒ Use DMR
☐ Set Temperature

[Download](#)

Run RP Analysis

Append DMR Data
[Download DMR Template](#)

Upload CSV to Append

[← Back to Search](#)

Data Overview

	Pollutant	Ref Param	# Samples	Unit_Status
1	pH (maximum)	00400	512	MATCH
2	pH (minimum)	00400	512	MATCH
3	Copper	01042	167	FAIL (Partial)
4	Temperature (summer)	00010	111	MATCH
5	Temperature (winter)	00010	111	MATCH
6	Cyanide	51173	109	MATCH
7	Zinc	01092	84	PASS
8	Silver	01077	58	MATCH
9	Thallium	01059	56	MATCH
10	Mercury	71900	52	MATCH
11	Nickel	01067	5	MATCH
12	Pentachlorophenol	39032	4	MATCH
13	Arsenic	01002	2	MATCH
14	Cadmium	01027	2	MATCH
15	1,1,1-trichloroethane	34506	0	NO DATA
16	1,1,2,2-tetrachloroethane	34516	0	NO DATA
17	1,1,2-trichloroethane	34511	0	NO DATA
18	1,1-dichloroethylene	34501	0	NO DATA
19	1,2,4-trichlorobenzene	34551	0	NO DATA
20	1,2-dichlorobenzene	34536	0	NO DATA
21	1,2-dichloroethane	32103	0	NO DATA
22	1,2-dichloropropane	34541	0	NO DATA
23	1,2-diphenylhydrazine	34346	0	NO DATA
24	1,2-trans-dichloroethylene	34546	0	NO DATA

Figure 2: Summary page showing the Coverage Summary table with unit status column

Understanding the Coverage Summary columns:

Column	Description
Pollutant	NPDES pollutant name from the form crosswalk

Column	Description
Form	The NPDES application form(s) that require this pollutant
Ref Param	The DMR parameter code(s) matched to this pollutant in the crosswalk
# Samples	Count of qualifying DMR records retrieved for this permit and date range
Unit Status	Result of the unit conversion check (see below)

Unit Status values:

Status	Meaning	Action
MATCH	Reported units already match the WQS unit (including recognized aliases) — no conversion needed	None required
PASS	Units differed but a multiplicative or temperature conversion was successfully applied	Conversion is shown in the report
FAIL (Partial)	Some eligible records converted successfully but others could not	Proceed with caution; non-convertible records are dropped from the RP calculation
FAIL	No eligible record could be converted to the WQS unit (unrecognized unit)	Investigate the DMR data; values are retained at face value but dropped from RP
EXCLUDED	Every record is a mass-load or flow unit (e.g., kg/d, MGD)	Informational — these cannot be compared to a concentration criterion and are excluded by design
NO WQS	No WQS criterion exists for this parameter in the selected water class	Informational — see the Needs Attention tab to add a manual criterion if appropriate
NO DATA	Parameter is in the crosswalk but had no DMR records	Informational — nothing to evaluate

EXCLUDED, NO WQS, and NO DATA are informational, not errors; they are color-coded distinctly (grey / blue-grey / light-grey) from the FAIL states (red / orange). A FAIL or FAIL (Partial) status

does not prevent the analysis from running, but it means that some or all values for that parameter may be in the wrong units. Before proceeding, verify the DMR data in the source ICIS-NPDES record for any parameters showing a FAIL status — the issue is often a data entry error in the original DMR submission.

4.5 Step 4b: Resolve Parameters Needing Attention

The Summary page has three tabs: **Data Overview** (the Coverage Summary described above), **[!] Needs Attention**, and **WQS Reference**.

The **Needs Attention** tab lists any parameter that has DMR data but no usable WQS criterion for the selected water class. Each is shown with the reason it was flagged:

- **Wrong water class** — a WQS criterion exists for this parameter, but not for the class(es) you selected.
- **No crosswalk entry** — the parameter code has no WQS entry at all.
- **Missing criterion value** — a crosswalk entry exists but its value is blank.

For each flagged parameter you choose **Include** or **Exclude**. To include a parameter, enter a criterion value, select a unit (limited to units the converter can handle), choose the applicable water class(es), pick a criterion type, and add a basis note. Included parameters are injected as manual criteria when you run the analysis; excluded parameters are recorded in the report's *Excluded Parameters* section for the permanent record. Use the **WQS Reference** tab — a read-only, searchable view of the full crosswalk — to look up criterion IDs and values before entering a manual criterion.

! Important

Run RP is gated on resolution. The **Run RP Analysis** button stays disabled until every flagged parameter has an Include/Exclude decision. If you need to proceed without resolving them, use the red **[!] Quick Run (Drop Unresolved)** button — it auto-excludes all unresolved parameters and stamps a prominent “preliminary report” warning on the generated report so the omission is on the permanent record.

Parameters that don't require resolution are never flagged: hardness-dependent metals and TAN (whose criteria are computed at run time from your hardness, pH, and temperature inputs) are treated as matched, and parameters reported only in mass-load or flow units are dropped from the flag pass because they can never be compared to a concentration criterion.

4.6 Step 5: Configure Analysis Settings

The left panel of the Summary page contains the Analysis Settings. These control the statistical and receiving water parameters used in the RWC and RP calculations. A full description of the

methodology behind each setting is provided in Section 8.

4.6.1 Dilution Ratio

The **Dilution Ratio** (DR) is the ratio of the total flow at the edge of the mixing zone (effluent plus receiving water) to the effluent flow alone. It represents the degree to which the discharge is diluted by the receiving water at the point of compliance. The default value is 1, which means no dilution is assumed — the effluent concentration itself is compared to the WQS criterion. This is the conservative default and is appropriate when:

- No mixing zone has been authorized
- Dilution data are unavailable
- The receiving water is intermittent or has negligible flow relative to the discharge

If a mixing zone authorization exists or if a dilution analysis has been performed, enter the appropriate dilution ratio here. Values greater than 1 reduce the RWC and make an RP finding less likely. The dilution ratio is captured in the report.

Warning

Dilution ratio and mixing zones: A dilution ratio greater than 1 implies that a mixing zone exists. Ensure that any mixing zone used here is either already authorized by DNER under Rule 1305 of the PRWQSR, or that you have an appropriate basis for the dilution value. Using an unsupported dilution ratio in the RP analysis may underestimate RP and result in a permit that does not adequately protect water quality.

4.6.2 Confidence Level and Target Percentile

These two parameters control the statistical projection used to compute the Multiplying Factor (MF) — see Section 8.1.1 for a full explanation.

Confidence Level (CL): The probability (expressed as a proportion between 0 and 1) that the highest observed value in the sample set exceeds the stated percentile of the underlying effluent distribution. The default is 0.95 (95%). EPA’s TSD presents lookup tables at both 95% and 99% confidence levels; the 95% level is the default used in most regional RP guidance.

Target Percentile: The upper bound of the effluent distribution that the projected RWC is designed to represent, expressed as a proportion. The default is 0.95 (95th percentile). Together with the confidence level, this defines how conservatively the projection treats variability in the effluent — a higher target percentile produces a larger MF and a more conservative RP determination.

i Note

Default settings: The defaults of 95% confidence level and 95th percentile target are consistent with EPA Region 2 guidance for RP determinations using existing DMR data. If regional or program-specific guidance specifies different values (e.g., 99%/99% for certain pollutant categories), adjust these settings accordingly. The values used are documented in the report.

4.6.3 Hardness Settings (Class SD only)

This panel appears only when **Class SD** is selected as a water classification. It controls how the app handles the seven hardness-dependent metal criteria in the PRWQSR (see Section 8.2).

Hardness value (mg/L as CaCO₃): Enter a numeric hardness value for the receiving water if a measured or estimated value is available. The app uses this value to compute the metal criterion from the hardness regression equations in Rule 1303.1(J)(1) and uses it directly in the RP comparison. The default is 100 mg/L as CaCO₃, which is a reasonable midrange value for Puerto Rico surface waters in the absence of site-specific data; however, you should use a measured value whenever possible.

4.6.4 pH and Temperature (for TAN, Class SD only)

These inputs appear within the Class SD conditional panel and are used only for the Total Ammonia Nitrogen (TAN) RP evaluation (see Section 8.3). The TAN criterion under Rule 1303.2(C)(2)(1) is a function of receiving water pH and temperature.

Use DMR data: If pH (parameter code 00400) and temperature (parameter code 00010) data are available in the DMR records retrieved from the database, the app will use those values to compute a time-varying TAN limit — one for each monitoring period that has both a pH and temperature record. The RP determination for TAN is YES if the TAN RWC exceeds the criterion at any point in the monitoring record (the “ever crossed” rule). The mean TAN limit is shown in summary tables and plots for display purposes.

Override with fixed values: If the DMR does not include pH or temperature, or if you prefer to use design receiving water conditions, enter fixed pH and temperature values. The app will use these constant values to compute a single TAN limit for comparison against the TAN RWC.

Receiving Water pH (SU): Valid range 0–14. Default is 6.0 SU, which represents a slightly acidic condition and produces a more conservative (lower) TAN criterion. For context, Class SD waters have a pH standard of 6.0–9.0 SU (Rule 1303.2(C)(2)(d)).

Receiving Water Temperature (°C): Valid range 0–50°C. Default is 30°C, consistent with Puerto Rico’s thermal discharge standard of 30°C (Rule 1303.1(D)(1)).

4.7 Step 6: Append Local DMR Data (Optional)

If you have effluent monitoring data that are not available in the database — for example, data from a recent sampling event that has not yet been submitted to EPA, or data from a non-standard monitoring program — you can append those records to the retrieved dataset before running the RP analysis.

Download the DMR Template: Click **Download DMR Template** to download a CSV file pre-populated with the parameter codes and expected WQS units for the selected facility and forms. The template includes one row per parameter code with blank value fields.

Complete the template: Fill in the following columns for each record you want to add:

Column	Description
<code>parameter_code</code>	5-digit DMR parameter code (pre-filled from crosswalk)
<code>parameter_desc</code>	Parameter description (informational only)
<code>dmr_value_nmbr</code>	Observed concentration (numeric)
<code>dmr_unit_desc</code>	Units for the reported value
<code>monitoring_period_end_date</code>	End date of the monitoring period (YYYY-MM-DD or MM/DD/YYYY)
<code>perm_feature_nmbr</code>	Outfall identifier (e.g., 001)

Multiple rows can be added for a single parameter code — add one row per monitoring period per outfall. Save the file as a CSV.

Upload the completed template: Click **Upload CSV to Append** and select your completed file. The app will validate the required columns, attempt unit conversion, and append the records to the existing DMR dataset. A notification will confirm the number of rows added.

Warning

Unit responsibility: When appending local data, you are responsible for ensuring that the units in `dmr_unit_desc` match what was actually measured. The app will attempt to convert to WQS units, but if the reported unit is not recognized, the value will be retained at face value and flagged with the appropriate unit status (e.g., FAIL, EXCLUDED, or NO WQS). Unit issues in appended data will appear in the Coverage Summary table.

4.8 Step 7: Run the RP Analysis

Click **Run RP Analysis**. (This button is enabled once any flagged parameters have been resolved — see Section 4.5.) The app computes the RWC and RP determination for every parameter ×

outfall combination in the dataset using the settings configured in Step 5. When the calculation is complete, the app navigates to the **Findings page**.

5 Standalone Workflow

Use the Standalone Calculator when no NPDES ID exists (e.g., for a new permit application) or when you have local monitoring data and prefer not to use the database-backed lookup.

5.1 Step 1: Open the Standalone Calculator

Click **Standalone Calculator** on the landing page. This opens the standalone analysis page with the same form and water classification selectors described in Section 4.2. Configure these selectors first — they define which parameters are expected and which WQS criteria will be used.

5.2 Step 2: Download and Complete the DMR Template

Click **Download DMR Template**. Because no facility has been selected, the template is generated from the crosswalk based solely on the selected NPDES forms and water classifications. It will contain one row per expected parameter code.

Complete the template as described in Section 4.7, including the `perm_feature_nmbr` (outfall) column. If the facility has multiple outfalls, add separate rows for each outfall for each parameter.

5.3 Step 3: Upload and Configure

Click **Upload Completed DMR Template** and select your CSV file. The app will validate the required columns, apply unit conversions, and display the Coverage Summary table. Review the unit status as described in Section 4.4.

Configure the **Analysis Settings** — dilution ratio, confidence level, target percentile, and (if Class SD is selected) the hardness and pH/temperature settings — exactly as described in Section 4.6.

5.4 Step 4: Run the RP Analysis

Click **Run RP Analysis** to proceed to the Findings page, where the results are presented in the same format as the NPDES workflow.

6 Findings Page

The Findings page is the primary results view. It is organized into a left sidebar with controls and summary statistics, and a main panel with interactive visualizations and the RP results table.

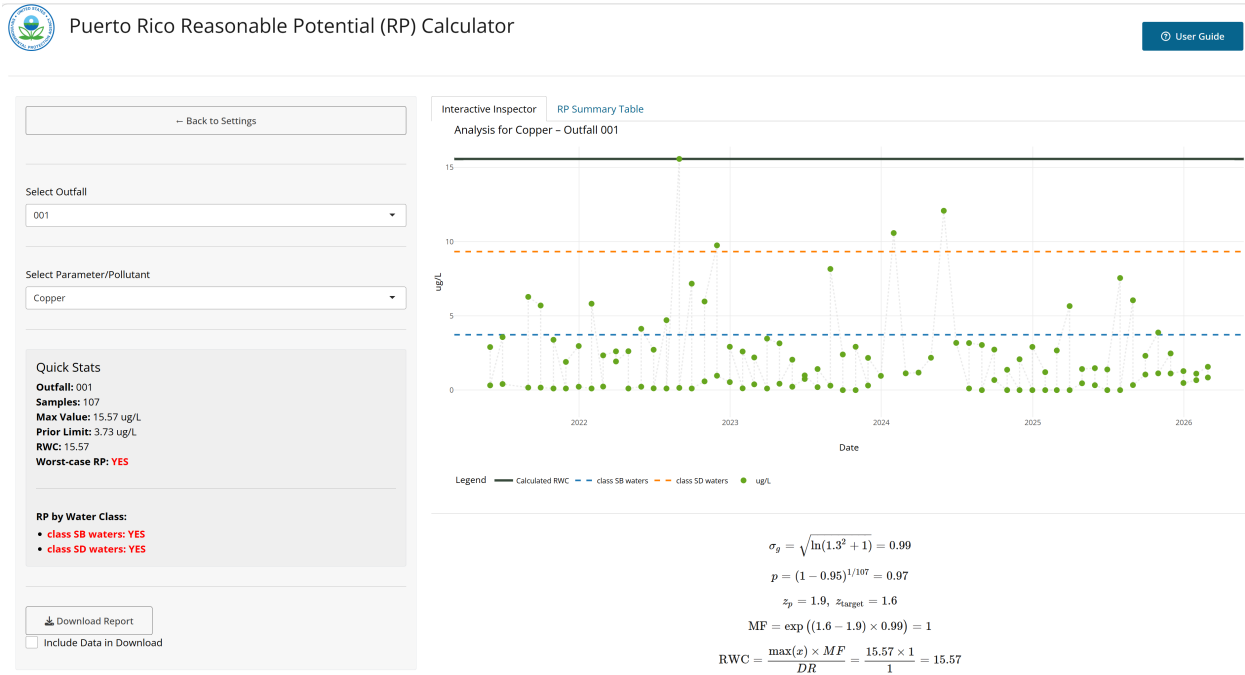


Figure 3: Findings page overview showing sidebar with outfall/pollutant selectors and Quick Stats panel, and main tabbed plot area

6.1 Outfall and Pollutant Selection

Use the **Select Outfall** and **Select Parameter/Pollutant** dropdowns in the left sidebar to navigate the results. The pollutant list is filtered to show only pollutants that have at least one observation for the selected outfall.

6.2 Quick Stats Panel

The Quick Stats panel in the left sidebar provides a summary of the key values for the selected outfall and pollutant:

Field	Description
Outfall	The currently selected outfall identifier
Samples	Count of observations included in the RWC calculation for this outfall × pollutant

Field	Description
Max Value	The highest observed concentration (in WQS units) — this is the value multiplied by the MF in the RWC formula
Prior Limit	The most recent numeric effluent limit present in the DMR record for this pollutant and outfall, if any (sourced from the DMR <code>limit_value_nmbr</code> field — not app-calculated)
RWC	The calculated Receiving Water Concentration = (Max Value × MF) ÷ Dilution Ratio
Worst-case RP	The headline result across selected classes: YES if any class is YES, otherwise NO. When a class resolves to DEPENDS (a hardness-dependent metal in hardness-range mode), that class is shown in the per-class breakdown below, with a note pointing to the Hardness vs Limit tab.

When multiple water classifications are selected, the panel also shows a per-class RP breakdown listing the RP result for each class independently.

RP result values:

Result	Meaning
YES	The RWC reaches or exceeds the applicable criterion for this water class (the comparison is $\geq$, so a value exactly equal to the criterion is an exceedance). Reasonable potential exists.
NO	The RWC is below every applicable criterion for this water class.
DEPENDS	A hardness-dependent metal was evaluated in hardness-range mode, so the result depends on the receiving-water hardness. Resolve by entering a site-specific hardness value, or read the critical-hardness threshold from the Hardness vs Limit plot.

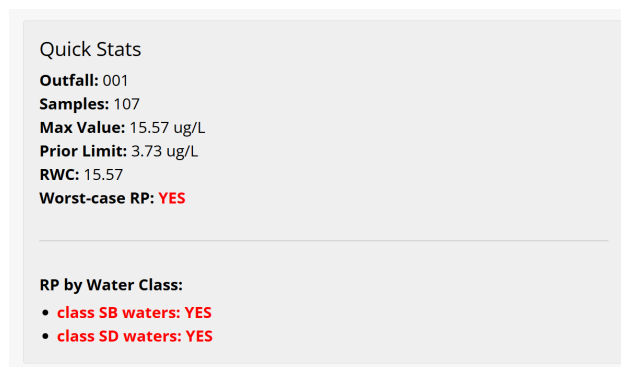


Figure 4: Quick Stats panel showing Worst-case RP highlighted in red (YES), with per-class breakdown

6.3 Interactive Inspector

The main panel shows the Observed vs. WQS/RWC plot. For hardness-dependent metals under Class SD evaluated in hardness-range mode, a second tab (“Hardness vs Limit”) is added alongside it.

6.3.1 Observed vs. WQS / RWC Plot

The primary plot shows the effluent monitoring data in the context of the applicable WQS criteria and the calculated RWC.

- **Points:** Individual DMR observations over time, colored by reported unit. Hovering over a point shows the value, date, and any limit value from the DMR record (prior permit limit, if available).
- **Dashed horizontal lines:** The WQS criterion for each selected water classification. When multiple classes are selected, each class appears as a separate dashed line labeled with the class name.
- **Solid horizontal line:** The calculated RWC for this outfall and parameter.
- **Y-axis:** Constrained to zero or greater. Very high RWC values relative to the criterion will cause the WQS lines to appear near the x-axis — this is expected.

For **hardness-dependent metals** when the range analysis option is active: an annotation is displayed on the plot reading “RWC would exceed WQS at Hardness < X mg/L as CaCO₃,” where X is the critical hardness threshold calculated from the regression equation (see Section 8.2). This tells you the receiving water hardness below which reasonable potential would exist.

For **Total Ammonia Nitrogen**: if time-varying TAN limits are being used (i.e., pH and temperature from the DMR), an additional line labeled “TAN Limit (variable)” is shown on the plot. This line represents the TAN criterion computed for each monitoring period using the co-occurring pH and

temperature values. The RP determination is based on whether the TAN RWC ever crosses this variable limit line.

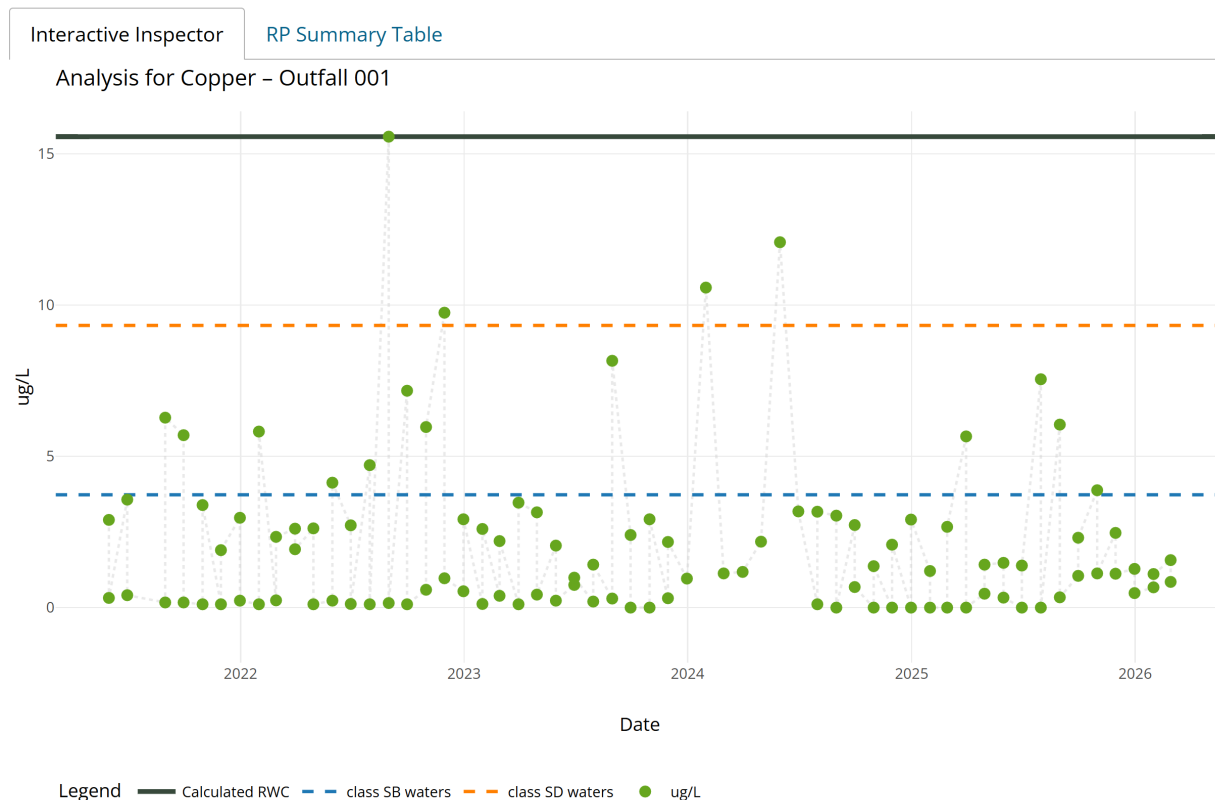


Figure 5: Observed vs. WQS/RWC plot with multiple WQS dashed lines, solid RWC line, and sample points colored by unit

6.4 RP Summary Table

The **RP Summary Table** tab in the main panel shows the full RP results for the selected outfall in tabular form, with one row per pollutant $\times$ WQS criterion $\times$ water class combination. Columns include the outfall, WQS criterion ID, pollutant name, criterion value, unit, water class, RWC, and RP determination. Rows are ordered by RP result (YES $\rightarrow$ DEPENDS $\rightarrow$ NO) to highlight the most critical findings, and the RP cell is color-coded (red for YES, orange for DEPENDS, green for NO). The table also supports per-column sorting and filtering. Where a Class SD sub-class applies, the Water Class column shows the combined label (e.g., *class SD waters — stream*), so stream and reservoir/lake criteria appear as separate rows. Clicking a row syncs the sidebar outfall and pollutant selectors and the math panel to that row.

Interactive Inspector

RP Summary Table

Show 15 entries

Search:

Outfall	WQS ID	Pollutant	WQS	Unit	Water Class	RWC	RP
001	79235	Copper	3.73	ug/L	class SB waters	15.57	YES
001	79236	Copper	9.33	ug/L	class SD waters	15.57	YES
001	79232	Cyanide	1	ug/L	class SB waters	1.9	YES
001	79244	Mercury	0.05	ug/L	class SB waters	0.1	YES
001	79245	Mercury	0.05	ug/L	class SD waters	0.1	YES
001	79247	Nickel	8.28	ug/L	class SB waters	100	YES
001	79248	Nickel	46.41	ug/L	class SD waters	100	YES
001	79261	Thallium	0.24	ug/L	class SD waters	0.42	YES
001	79224	Arsenic	36	ug/L	class SB waters	6.45	NO
001	79225	Arsenic	10	ug/L	class SD waters- drinking water	6.45	NO
001	79227	Cadmium	7.95	ug/L	class SB waters	0.27	NO
001	79228	Cadmium	0.79	ug/L	class SD waters	0.27	NO
001	79323	Pentachlorophenol	0.4	ug/L	class SB waters	0	NO

Figure 6: RP Summary Table with YES rows highlighted at top, showing all columns

7 Report Generation

7.1 Download Options

Click **Download Report** in the left sidebar of the Findings page. Two download formats are available:

PDF only (default): A single PDF report rendered from a Quarto template. The report documents all analysis inputs, settings, and findings in a format suitable for inclusion in a permit record.

PDF + Data (ZIP): Check the **Include Data in Download** checkbox before clicking Download Report. The app will bundle the PDF together with:

- **data.xlsx** — an Excel workbook with five sheets: Summary Findings (key RP results filtered to outfalls with $RWC > 0$), RP_Concentration (full results table), DMR (all effluent records

used in the analysis), Coverage (the NPDES forms coverage table), and WQS_Info (the effective WQS criteria actually used, including run-time-calculated metal and TAN values)

- **README.txt** — a short text file noting the permit ID, the queried date range, and the hardness setting

Use the ZIP option when you want to archive the underlying data alongside the report, or when the analysis findings will be reviewed by others who need access to the raw numbers.

7.2 Report Contents

The PDF report includes:

Summary section: Permit ID, facility name, date range, NPDES forms, water classifications, dilution ratio, confidence level, target percentile, hardness setting, and the number of outfalls analyzed.

Per-pollutant sections: One section per pollutant per outfall (when multiple outfalls exist). Each section includes:

- The observed vs. WQS/RWC plot
- The RP determination and key statistics (n, max value, MF, RWC, criterion)
- A limit history table if prior permit limits were present in the DMR record
- For hardness-dependent metals with range analysis: both the hardness vs. limit plot and the observed vs. WQS/RWC plot with the critical hardness annotation
- For TAN: the variable limit line in the plot and a note on the mean TAN limit used for display

Coverage table: The NPDES forms coverage summary showing which expected pollutants had data and the unit conversion status for each. Manually-entered criteria are flagged with a † marker and footnote.

Conditional sections: Depending on the analysis, the report may also include:

- **Quick Run warning** (if Quick Run was used) — a prominent notice listing the parameters that were auto-excluded, flagging the report as preliminary.
- **Reason Included line** under each pollutant heading — whether the pollutant was pulled in by a selected form or included because prior DMR data exists.
- **Manual WQS callouts** — an in-line callout in a pollutant's RP subsection when a manually-entered criterion was applied.
- **Excluded Parameters** (if any were excluded) — a permanent record of the parameters that were dropped and the stated reasons.
- **Manual WQS Overrides** (if any were entered) — a full audit table of every manually-entered criterion: value, unit, water class, basis note, and the original flag reason.
- **Unavailable Pollutants** — pollutants expected but without usable data.

i Note

Report generation time: Rendering the PDF involves running a Quarto document with embedded plots for every pollutant × outfall combination. For facilities with many parameters and multiple outfalls, this can take 30–90 seconds. A progress indicator is displayed during rendering. Do not close the browser tab while the report is generating.

8 Methodology

This section describes the statistical and regulatory methods used by the app to calculate RWC values, derive RP multiplying factors, and evaluate special cases including hardness-dependent metals, Total Ammonia Nitrogen, pH, and temperature. It is intended to provide a transparent account of what the app computes and why.

8.1 The Statistical Projection Approach (TSD Section 3.3.2)

The fundamental question in an RP determination is whether the effluent, after accounting for dilution, could cause an exceedance of the applicable criterion in the receiving water. The challenge is that effluent concentrations are variable — a single measured maximum may not represent the true upper bound of what the discharge is capable of producing.

EPA’s TSD addresses this by treating the effluent concentration data as a sample from an underlying lognormal distribution and projecting the upper bound of that distribution at a specified confidence level. This approach does two things simultaneously:

1. It accounts for **data uncertainty** — the fact that a small number of samples represents the highest value less reliably than a large number of samples does
2. It estimates the **statistical upper bound** of the effluent distribution at a user-specified probability level (the target percentile)

The result is a Multiplying Factor (MF) that is applied to the observed maximum concentration before comparing to the criterion.

8.1.1 The Multiplying Factor

The MF is derived in two steps following Box 3-2 of the TSD.

Step 1 — Percentile represented by the maximum observation:

For a dataset of n samples at a given confidence level CL, the highest observed value exceeds the p_n -th percentile of the underlying distribution with probability CL, where:

$$p_n = (1 - CL)^{1/n}$$

For example, with $n = 10$ samples and $CL = 0.95$, the maximum value exceeds approximately the 74th percentile of the distribution with 95% confidence.

Step 2 — Project from the observed percentile to the target percentile:

Assuming a lognormal distribution for the effluent, the ratio between the target percentile and the observed percentile can be expressed as:

$$MF = \exp \left[(z_{target} - z_p) \cdot \sigma_g \right]$$

where z_{target} is the standard normal variate corresponding to the target percentile, z_p is the standard normal variate corresponding to p_n , and $\sigma_g = \sqrt{\ln(CV^2 + 1)}$ is the geometric standard deviation of the lognormal distribution expressed in terms of the coefficient of variation (CV).

The **coefficient of variation** is estimated as:

$$CV = \frac{s}{\bar{x}}$$

where s is the sample standard deviation and $\bar{x}$ is the sample mean. For datasets with fewer than 10 samples, the CV cannot be estimated reliably from the data alone. In this case, the app assigns a default CV of **0.6**, consistent with the TSD recommendation and EPA's general characterization of effluent variability based on its national database.

The MF is bounded to be no less than 1. An MF below 1 would imply that the maximum observed value already exceeds the target percentile — in this case no upward projection is needed, and the maximum value itself is used.

8.1.2 The Receiving Water Concentration

The RWC is calculated as:

$$RWC = \frac{C_{max} \times MF}{DR}$$

where C_{max} is the maximum observed effluent concentration (in WQS units), MF is the multiplying factor from the statistical projection, and DR is the dilution ratio. The RP finding is YES if $RWC \geq$ criterion for any applicable water class (a value exactly equal to the criterion is treated as an exceedance).

8.1.3 Worked Example

Consider a facility with 8 observations of a pollutant at an outfall, with a maximum of 45 µg/L, a mean of 18 µg/L, and a standard deviation of 12 µg/L. Analysis settings are CL = 0.95, target percentile = 0.95, and DR = 1. Assume the criterion for the receiving water is 36 µg/L (aquatic life protection).

$$\begin{aligned}
 CV &= 12/18 = 0.667 \\
 \sigma_g &= \sqrt{\ln(0.667^2 + 1)} = \sqrt{\ln(1.444)} = 0.603 \\
 p_n &= (1 - 0.95)^{1/8} = 0.05^{0.125} = 0.693 \\
 z_p &= \Phi^{-1}(0.693) = 0.503 \\
 z_{target} &= \Phi^{-1}(0.95) = 1.645 \\
 MF &= \exp[(1.645 - 0.503) \times 0.603] = \exp(0.689) = 1.99 \\
 RWC &= (45 \times 1.99)/1 = 89.5 \text{ µg/L}
 \end{aligned}$$

Since RWC (89.5 µg/L) ≥ criterion (36 µg/L), the RP determination is **YES**.

The app displays each of these intermediate values — CV, σ_g , p_n , z_p , z_{target} , MF, and RWC — in the math inspector panel on the Interactive Inspector tab, providing a fully transparent step-by-step calculation.

8.2 Hardness-Dependent Metal Criteria

Seven metals in Class SD waters have criteria that depend on receiving water hardness under Rule 1303.1(J)(1) of the PRWQSR. The criterion is a function of the natural logarithm of hardness expressed in mg/L as CaCO₃:

$$\text{Criterion} = e^{(LC \cdot \ln(H) - RC)}$$

where H is the receiving water hardness and LC and RC are regression constants specific to each metal. The constants used in the app are taken directly from the PRWQSR footnotes to Table 1 in Rule 1303.1(J)(1):

Metal	CRITERION_ID	LC	RC
Cadmium	79228	0.7977	3.909
Chromium III	79238	0.8190	0.6848
Copper	79236	0.8545	1.702
Lead	79253	1.273	4.705

Metal	CRITERION_ID	LC	RC
Nickel	79248	0.8460	0.0584
Silver	79251	1.72	6.59
Zinc	79264	0.8473	0.884

When a specific hardness value is entered in the Analysis Settings, the app computes the criterion at that hardness and uses it directly in the RP comparison. When the range analysis option is active, the app computes the **critical hardness** — the receiving water hardness at which the criterion would equal the RWC — by inverting the formula:

$$H_{critical} = e^{(\ln(RWC)+RC)/LC}$$

If $H_{critical}$ is below the plausible range for the receiving water body (e.g., below 25 mg/L as CaCO_3 , which is the lower end of the range plotted), reasonable potential is unlikely regardless of hardness. If $H_{critical}$ is within or above the plausible range, the permit writer should use a site-specific hardness value to resolve the determination.

8.3 Total Ammonia Nitrogen (TAN)

Total Ammonia Nitrogen in Class SD surface waters is subject to a criterion that varies with the pH and temperature of the receiving water body, as specified in Rule 1303.2(C)(2)(l) of the PRWQSR:

$$TAN = 0.8876 \times \left(\frac{0.0278}{1 + 10^{7.688 - pH}} + \frac{1.1994}{1 + 10^{pH - 7.688}} \right) \times (2.126 \times 10^{0.028 \times (20 - T)})$$

where T is the receiving water temperature in °C and the result is expressed as mg/L as N. This formula reflects the toxicity of un-ionized ammonia (NH_3) to aquatic life, which increases as pH rises and temperature rises. Because the TAN criterion is computed at run time from the receiving-water pH and temperature rather than stored as a fixed value, TAN is never flagged on the Needs Attention tab, and its calculated value is written into the exported WQS_Info data.

Parameter codes included: The app evaluates TAN using up to three DMR parameter codes that all represent total ammonia nitrogen with different reporting conventions:

Parameter Code	Description	Conversion to mg/L as N
00610	Nitrogen, ammonia total [as N]	None needed
71845	Nitrogen, ammonia total [as NH_4]	$\times 0.7778$
34726	Nitrogen, ammonia total [as NH_3]	$\times 0.8235$

Values reported as NH_4 or NH_3 are converted to the equivalent as-N concentration before the RWC is calculated.

Time-varying vs. fixed limits: When the app uses pH and temperature values from the DMR (parameter codes 00400 and 00010 respectively), the TAN criterion is computed separately for each monitoring period that has co-occurring pH and temperature observations. This produces a time-series of limits that reflects actual receiving water conditions. The RP determination is **YES** if the TAN RWC exceeds the criterion in *any* monitoring period (the “ever crossed” rule) — because a single exceedance represents a real potential for criterion violation. Summary tables display the **mean** TAN limit across all periods for readability.

When fixed pH and temperature overrides are used, a single constant TAN limit is computed and compared to the TAN RWC.

8.4 pH

The PRWQSR establishes pH ranges rather than absolute concentration limits. For Class SD, the applicable standard is 6.0–9.0 SU (Rule 1303.2(C)(2)(d)). For Class SB, the range is 7.3–8.5 SU (Rule 1303.2(B)(2)(d)). pH is now evaluated like any other parameter: when the selected water class has a pH criterion, the app compares the observed pH range to the standard — an observed maximum at or above the upper bound, or an observed minimum at or below the lower bound, yields a YES — and reports the result as separate *pH (maximum)* and *pH (minimum)* rows. This is a direct comparison of observed pH to the standard, not the statistical RWC projection, because pH is a range-based standard. pH values are also still used to compute the TAN criterion (see Section 8.3). If the selected class has no pH criterion, pH instead surfaces on the Needs Attention tab like any other unmatched parameter.

8.5 Temperature

The PRWQSR’s thermal discharge standard states that no heat shall be added to waters of Puerto Rico that would cause the temperature of any site to exceed 30°C or 86°F (Rule 1303.1(D)(1)). Temperature is now evaluated as a parameter: when the selected water class has the temperature criterion, the app compares the observed maximum temperature to the standard, and an observed maximum at or above the standard yields a YES, shown as a *Temperature* row. Temperatures reported in Fahrenheit are converted to Celsius before evaluation. As with pH, this is a direct comparison rather than the statistical RWC projection, and temperature data also support the TAN calculation.

9 Interpreting Results and Common Cases

9.1 Multiple Water Classes Selected

When more than one water classification is selected, the Observed vs. WQS/RWC plot shows a separate dashed criterion line for each class. The Quick Stats panel shows both the worst-case RP across all classes and a per-class breakdown. The RP Summary Table lists each class as a separate row for each pollutant, making it easy to see which classes are driving a YES finding and which are not.

A common scenario is selecting both Class SB and Class SD when a facility may discharge to either coastal or surface receiving waters depending on the outfall. In this case, the more protective of the two criteria will drive the worst-case RP result. Where a Class SD sub-class applies (total nitrogen, total phosphorus, or selenium), each receiving-water type — stream vs reservoir/lake — appears as its own criterion line and table row.

9.2 Metals with Hardness = Range (Class SD)

When the hardness range analysis is active for a hardness-dependent metal, the Quick Stats panel shows **Worst-case RP: DEPENDS**. To resolve this to a YES or NO, you have two options:

1. **Use a site-specific hardness value:** Obtain a measured or representative hardness value for the receiving water (from ambient monitoring data, the permit record, or DNER) and enter it in the Hardness setting. Re-run the analysis.
2. **Use the critical hardness threshold:** Read the critical hardness annotation from the plot or the math inspector. If the critical hardness is well above the expected range for the receiving water, a NO determination is defensible. Document the basis for the hardness assumption in the permit record.

9.3 Unit Conversion Issues

A FAIL or FAIL (Partial) unit status for a parameter means the app could not convert some or all reported values to the WQS unit. This does not stop the analysis — values are retained at face value — but the RWC for that parameter should be interpreted with caution. Common causes:

- **Unusual units in the DMR:** Some DMR submissions use non-standard unit abbreviations (e.g., “PPB” instead of “ug/L”) that the app does not recognize. If you can confirm the unit is equivalent, the face value is correct.
- **Genuine unit mismatch:** The permittee reported a value in a genuinely different concentration unit that the app does not recognize. In this case the face value is meaningless and the parameter should be flagged for investigation before the RP finding is relied upon.

Note that not every non-PASS status is a data problem. **EXCLUDED** means the parameter was reported in a mass-load or flow unit (e.g., kg/day) that cannot be compared to a concentration criterion — this is expected for load-based parameters, not an error. **NO WQS** means there is simply no criterion for the parameter in the selected class; use the Needs Attention tab to add one if appropriate. Only **FAIL** and **FAIL (Partial)** indicate an unrecognized unit that warrants investigation.

The unit conversion flag is documented in the report so that downstream reviewers can assess the data quality for affected parameters.

9.4 No Data for a Parameter

If the selected pollutant has no observations for the selected outfall, the Inspector plot will show a “no data” message and no row will appear in the RP Summary Table for that outfall × pollutant combination. This can occur when:

- The parameter was not monitored at that outfall during the selected date range
- The parameter code in the crosswalk does not match the code used in the DMR (check the Coverage Summary)
- The DMR data were submitted under a different outfall identifier

If a parameter is required by the selected NPDES form but has no data, it will appear in the Coverage Summary with a **NO DATA** status and a sample count of 0. A parameter that does have data but no applicable criterion for the selected class shows a **NO WQS** status and appears on the Needs Attention tab. The report will include a note on parameters with missing data.

9.5 Prior Limits in the Quick Stats Panel

The **Prior Limit** field in Quick Stats shows the most recent numeric effluent limit from the DMR record for the selected parameter and outfall. This is sourced directly from the `limit_value_nmbr` and `limit_unit_desc` fields in the ICIS-NPDES DMR record — it reflects the limit as recorded in the prior permit, not a limit calculated by this app. It is provided as a reference point for permit writers comparing the current RP analysis to existing permit conditions.

10 Appendix A: NPDES Application Forms and Pollutants

The following table lists the pollutants included in the app’s crosswalk by NPDES application form. These are the pollutants for which WQS criteria exist in the PRWQSR and for which DMR parameter code matches have been established. Pollutants required by NPDES forms that do not

have a corresponding WQS criterion are tracked in the Coverage Summary; they are not evaluated for RP automatically, but a criterion can be supplied for them through the Needs Attention tab (see Section 4.5). The crosswalk also carries a `sub_class` column distinguishing stream vs reservoir/lake criteria under Class SD.

i Note

The full crosswalk table will be included here once the data file is linked. In the rendered application, this table is populated from the `npdes_forms` and `crosswalk` objects loaded at application startup.

11 Appendix B: Technical References

The following documents provide the regulatory and methodological basis for the RP Calculator.

EPA (1991). *Technical Support Document for Water Quality-Based Toxics Control*. EPA/505/2-90-001. Office of Water, U.S. Environmental Protection Agency, Washington, D.C. Section 3.3.2 — Statistical Projection of Upper-Bound Effluent Concentrations.

DNER (2025). *Puerto Rico Water Quality Standards Regulation, as Amended April 2025*. Resolution R-25-006. Puerto Rico Department of Natural and Environmental Resources, San Juan, PR. Filed with the Puerto Rico Department of State June 27, 2025 (Number 9677).

40 CFR 122.44(d)(1). *Conditions applicable to all permits — Technology-based and water quality-based effluent limitations*. Code of Federal Regulations, Title 40, Part 122.

40 CFR 131. *Water Quality Standards*. Code of Federal Regulations, Title 40, Part 131.

12 Appendix C: Glossary

Coefficient of Variation (CV): The ratio of the standard deviation to the mean for a dataset. Used in the MF calculation as a measure of relative effluent variability. When $n < 10$, the app assigns a default CV of 0.6 per TSD guidance.

Confidence Level (CL): The probability that the highest observed value in a sample of n exceeds a given percentile of the underlying distribution. Defaults to 0.95 (95%).

DEPENDS: An RP result indicating that the determination depends on receiving-water hardness. It is produced for hardness-dependent metals evaluated in hardness-range mode; resolve it by entering

a site-specific hardness value or by reading the critical-hardness threshold from the Hardness vs Limit plot.

DMR (Discharge Monitoring Report): The periodic report submitted by NPDES permittees documenting effluent monitoring results. Retrieved by the app from a local copy of EPA’s ICIS-NPDES database, refreshed weekly.

Dilution Ratio (DR): The ratio of the total receiving water flow (including the effluent) to the effluent flow alone at the point of compliance. Used to scale the RWC to receiving water conditions.

ECHO (Enforcement and Compliance History Online): EPA’s public database of NPDES permit and compliance data. The app does **not** query ECHO at run time; effluent data are read from a local copy of the ICIS-NPDES database. ECHO is used only in an offline validation script that cross-checks the local data.

ICIS-NPDES (Integrated Compliance Information System – NPDES): EPA’s national system of record for NPDES permit, facility, and DMR data. The app reads facility and effluent data from a local copy of ICIS-NPDES that is refreshed on a weekly schedule.

MF (Multiplying Factor): The statistical factor applied to the maximum observed effluent concentration to project the upper bound of the effluent distribution at the target percentile and confidence level. Bounded to be ≥ 1 .

NODI Code: A code in the DMR indicating a reason why no numeric value was reported for a monitoring period. NODI B and Q (no discharge) are treated as zero by the app; other NODI codes are excluded from the analysis.

NPDES (National Pollutant Discharge Elimination System): The federal permitting program under the Clean Water Act that regulates point source discharges to waters of the United States.

PRWQSR (Puerto Rico Water Quality Standards Regulation): The water quality standards regulation administered by DNER that establishes the numeric and narrative criteria protecting Puerto Rico’s waters. The 2025 amendment is the applicable version in this app.

Reasonable Potential (RP): Under 40 CFR 122.44(d)(1), the determination that a discharge causes, has the reasonable potential to cause, or contributes to an excursion above an applicable water quality standard. A YES RP finding triggers the requirement for a water quality-based effluent limitation in the permit.

RWC (Receiving Water Concentration): The projected concentration of a pollutant in the receiving water after accounting for the statistical upper bound of the effluent distribution and the available dilution: $RWC = (C_{\text{max}} \times MF) \div DR$.

Sub-class: A distinction within a water class whose WQS criterion differs by receiving-water type. In Puerto Rico’s Class SD waters, total nitrogen, total phosphorus, and selenium carry separate

stream and reservoir/lake criteria; the app shows these as combined labels such as *class SD waters — stream*.

TAN (Total Ammonia Nitrogen): The sum of ammonium ion (NH_4^+) and un-ionized ammonia (NH_3). Subject to a pH- and temperature-dependent criterion in Class SD waters under Rule 1303.2(C)(2)(1) of the PRWQSR.

Target Percentile: The upper bound of the lognormal effluent distribution that the projected RWC is designed to represent. Defaults to 0.95 (95th percentile).

TSD (Technical Support Document for Water Quality-Based Toxics Control): EPA's 1991 guidance document providing the statistical methodology for RP determinations using effluent monitoring data (EPA/505/2-90-001).

WQS (Water Quality Standards): Provisions of state or federal law that consist of a designated use or uses for a water body, the water quality criteria necessary to protect those uses, and an antidegradation policy. In this app, WQS refers specifically to the criteria established in the PRWQSR.